Putting Marginal Back in Tobin's q

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Overview

Question: marginal q drives investment, but it is unobservable

- q-theory: invest iff marginal $q > 1$, i.e., one more unit of capital is worth more than it costs (Tobin, 1969)
- The literature proxies it with **average** Q = market value / capital stock, a weak predictor of investment (Hayashi, 1982; Erickson & Whited, 2000; Philippon, 2009; Peters & Taylor, 2017)

Can we measure marginal q directly?

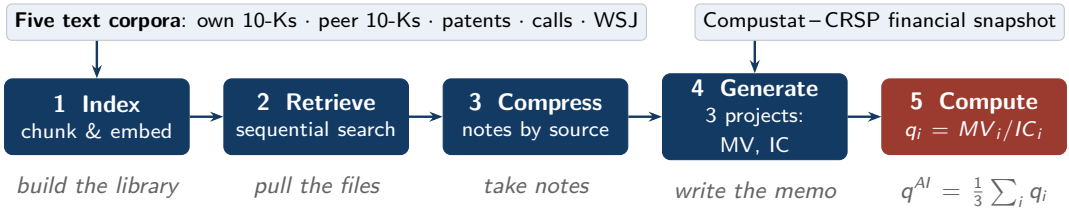
Approach: build q bottom-up from the marginal project

- An LLM reads retrieved firm text, proposes three likely next projects per firm-year, and values each; q^{AI} = average of value / cost

Findings

- q^{AI} predicts investment better than average Q; it also contains qualitatively distinct information about the composition of investment opportunities
- Average Q is inflated by omitted intangibles and market-power rents

Method: from firm text to marginal q



Design principles

- **No look-ahead**: to measure q in year t, the pipeline only uses documents dated before t
- The pipeline mimics an analyst: gather, read, write a memo

Step 1 Index: turn documents into a searchable library

WHAT HAPPENS

1. Split documents into short passages
A few sentences each; neighbours share a little text, so no idea is cut in half [overlap]

2. Turn each passage into numbers, by meaning
The model reads meaning, not exact words: the same idea in different words gets nearby vectors [embedding model: multilingual-e5-large; closeness = cosine similarity]

3. File it in the library
Each record keeps the vector, the text, and tags to filter by firm, year, source

VARIAN (RADIOTHERAPY MACHINES, CANCER-CARE SOFTWARE)

From Varian's 10-K, FY2006:
Passage 1: . . . Varian builds linear accelerators that cancer centers use for radiation therapy. We also plan to
Passage 2: We also plan to broaden our product line through new imaging tools and selective acquisitions. . .

Passage 1 → $v_1 = (0.12, -0.08, \ldots, 0.31)$ [1,024 numbers]
Rival: ". . . we supply radiotherapy equipment to hospitals. . . " → v_r: $\cos(v_1, v_r) = 0.91$, **close**
Varian: ". . . we lease our Palo Alto headquarters. . . " → v_h: $\cos(v_1, v_h) = 0.18$, **far**

vector: $(0.12, -0.08, \ldots, 0.31)$
text: ". . . Varian builds linear accelerators . . . We also plan to"
tags: Varian (gvkey 11115) · SIC 3845 · 2006 · 10-K business section

Step 2 Retrieve: search the library, expanding the question

1 Index → **2 Retrieve** → 3 Compress → 4 Generate → 5 Compute q

WHAT HAPPENS	VARIAN, FY2008
Embed the question Same embedding model as the library	q_0 = "What is Varian likely to invest in next?" → $u_0 = (0.05, 0.21, \ldots, -0.14)$
A. Search Varian's own 10-Ks Rank passages by cosine similarity to u_0; keep the top k [Varian only, dated before 2008]	**0.84** "...broaden our product line through acquisitions..." ✓ **0.79** "...image-guided therapy is a key growth area..." ✓ 0.18 "...we lease our Palo Alto headquarters..." ×
B. Expand the question, search rivals Append the hits and re-embed; same industry, Varian excluded	$q_1 = q_0 \oplus$ k own passages → u_1
C. Expand again: patents, calls, news The question now carries Varian's and its rivals' text	$q_2 = q_0 \oplus$ own $\oplus$ peer passages → u_2
Result: the evidence bundle	225 passages on Varian, all dated before FY2008

Step 3 Compress: turn the evidence into short notes

1 Index → 2 Retrieve → **3 Compress** → 4 Generate → 5 Compute q

What it does

- An LLM rewrites each source's passages into notes
- Two hard constraints:
  - **Faithful**: no facts beyond the source
  - **Dense**: keep dates, amounts, products, competitors
- One block per source, so sources do not mix

Varian, FY2008: compressed notes

Source	Key evidence
Own 10-K	Linear accelerators, treatment planning and data systems; grows via acquisitions, incl. a joint venture with Mitsubishi Electric
Peer 10-Ks	Fast-changing, competitive industry; heavy R&D; patents matter
Patents	X-ray tubes, plasma generation, image processing, ion beams
Earnings calls	**$30M ACCEL acquisition** (proton therapy); rollout of image-guided radiation therapy (IGRT); cargo screening, sales $\approx$ \$10M
WSJ	Makes in-house products once imported from Japan and Korea

Steps 4 – 5 Generate projects and compute q^{AI}

1 Index → 2 Retrieve → 3 Compress → **4 Generate** → **5 Compute q**

Input

- Notes from the five sources
- Financials: assets, capex, R&D, market cap, revenue

Output: exactly 3 projects

- Each with description, **market value**, **implementation cost** , reasoning

Compute

- $q_i = MV_i / IC_i$; q^{AI} = mean

Sample: 455 firms with ≥ 25 years of data, 1994–2022: 10,616 firm-years

Varian, FY2008: model output

Project	Main evidence	q_i
1. Proton therapy expansion	Calls: \$30M ACCEL acquisition	3.50
2. IGRT software suite	Patents (image processing) + calls (IGRT)	3.73
3. Cargo screening	Calls ($\approx$\$10M sales) + WSJ	3.88
q^{AI} = average of the three projects		**3.70**

Project 1 in detail: proton therapy system expansion

Description: scale up production and service of the proton-therapy systems from the 2007 ACCEL acquisition; cut installation times and improve reliability

Reasoning: earnings calls show strong interest in proton therapy; Varian already has a foothold; fits its focus on oncology and whole-system delivery

Labels: physical · buy · low compatibility

q^{AI} explains investment better than average Q

$$I_{it}^{tot}/K_{i,t-1}^{tot} = \beta\, q_{i,t-1} + \alpha_i + \eta_t + \varepsilon_{it}, \qquad q \in \{Q^{std},\ Q^{tot},\ q^{AI}\}$$

Table 3: Total investment rate on lagged q

	(1)	(2)	(3)
L.Q^{std}	0.012		
	(0.009)		
L.Q^{tot}		1.349***	
		(0.184)	
L.q^{AI}			**2.552*****
			(0.148)
Firm & year FE	Y	Y	Y
Observations	10,616	10,616	10,616
$\Delta R^2_{\text{within}}$: Q^{tot} vs. Q^{std}		0.168***	(0.022)
$\Delta R^2_{\text{within}}$: q^{AI} vs. Q^{std}		0.209***	(0.024)
$\Delta R^2_{\text{within}}$: q^{AI} vs. Q^{tot}		**0.042*****	(0.015)

Benchmarks

$Q^{std} = V/K^{phys}$,
$Q^{tot} = V/(K^{phys} + K^{int})$

Coefficient

Q^{std} 0.012 (insignificant) $\rightarrow$ Q^{tot} 1.349 $\rightarrow$ q^{AI} 2.552

Within-firm R^2

q^{AI} adds 0.209 over Q^{std} and **0.042** over Q^{tot}: modest but significant information in text that accounting aggregates miss

q^{AI} beats both benchmarks: a larger coefficient and a better within-firm fit

Investment responds more when the project matches its type

$$I^j_{it}/K_{i,t-1} = \beta_0\, q^{AI}_{i,t-1} + \beta_1\, \mathbf{1}(\text{Type } j)_{i,t-1} + \beta_2\, q^{AI}_{i,t-1} \times \mathbf{1}(\text{Type } j)_{i,t-1} + \alpha_i + \eta_t + \varepsilon_{it}$$

Table 6: Heterogeneity by project type

	Physical	Intangible	R&D
L.q^{AI}	0.854***	1.435***	0.883***
	(0.025)	(0.059)	(0.026)
L.$\mathbf{1}$(Type)	−0.110	−0.069	−0.261**
	(0.100)	(0.154)	(0.119)
L.$q^{AI} \times \mathbf{1}$(Type)	**0.132*** **	**0.145** **	**0.228*** **
	(0.043)	(0.064)	(0.050)
Firm & year FE	Y	Y	Y
Observations	10,616	10,616	10,616
R^2	0.537	0.730	0.765

Elasticity when the project matches

- R&D: 0.883 → 1.111 (**+26%**)
- Physical: 0.854 → 0.986 (+15%)
- Intangible: 1.435 → 1.580 (+10%)

q^{AI} is not just a better-measured Q – it also identifies the nature of the marginal project

Why average Q fails: intangibles, rents and frictions affect the wedge

$$Q^{tot}_{i,t-1} - q^{AI}_{i,t-1} = \gamma' X_{it} + \alpha_i + \eta_t + \varepsilon_{it}$$

Table 9: Dissecting the wedge, column (5)

Channel	Proxy	Coefficient
Omitted intangibles	Intangible intensity	**0.127*****
		(0.012)
Market-power rents	Profit margin	**0.053*****
		(0.012)
Financial frictions	WW index	−2.282***
		(0.327)
	Book leverage	−0.018
		(0.781)
Scale	$\log K^{tot}$	−1.310***
		(0.265)
Firm & year FE; R^2		Yes; 0.715

- **Intangibles** (largest): Q's denominator misses knowledge and organization capital
- **Rents**: pricing power raises market value, not the return on the next project
- **Frictions**: constrained firms have *smaller* wedges; constraints depress Q, while q^{AI} still values the projects

Discussion

My overall takeaway

A very interesting idea

- A fresh angle: build q bottom-up from the marginal project, not top-down from average Q
- q becomes an object: project type and fit with existing operations enable new tests
- A portable framework: LLMs construct, not just extract, latent economic objects

My concerns

- **Is q^{AI} computed correctly?**
  Does it value projects from the text, or does it pick up Q^{tot}, the LLM's memory, and lagged financials?
- **Does q^{AI} truly measure marginal q?**
  Predicting investment (a quantity) vs. measuring a shadow value (a price)

Comment 1.1: How much of q^{AI} comes from the projects?

- Financials enter the prompt by design, so co-movement with Q^{tot} is expected
- But the quantiles almost coincide, and Varian's three very different projects all sit near $Q^{tot} = 4.00$ (3.50 / 3.73 / 3.88)
- The LLM may be anchoring each project on the firm's market-to-book:
  - The "bottom-up" content would be thin
  - The wedge $Q^{tot} - q^{AI}$ become partly mechanical
- SFT targets Q^{tot} directly; which version enters Tables 3–9 is unclear

Distribution of q measures (Table 2, $N = 10{,}616$)

	25%	50%	75%	Mean	SD
Q^{tot}	**0.580**	**1.043**	**1.816**	1.645	2.480
q^{AI}	**0.587**	**1.039**	**1.798**	1.562	1.750

SFT (optional, Section 3.5)

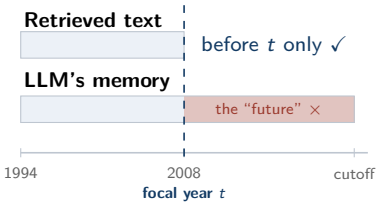
1. **Raw output**: run the RAG pipeline on all firm-years
2. **Pseudo-labels**: keep project text; set IC = total investment, MV = Q^{tot} × IC
3. **Train**: fine-tune the LLM (QLoRA) to map each prompt to its pseudo-label; 80/10/10 split by firm
4. **Re-generate**: run the fine-tuned model on the same prompts

Suggestions

- Perturb market cap in the snapshot while holding the text fixed: does q^{AI} move?
- Report results with pre-SFT and post-SFT q^{AI} separately

Comment 1.2: The date filter covers the text, not the model's memory

- Retrieval uses only pre-t text, but the LLM's memory runs to its training cutoff: it may know what Varian did after 2008
- Paper's defence: grounding, near-zero temperature, anonymization ("robust")
- Why this may not be enough:
  - The model and its cutoff are not reported
  - Anonymization hides the name, not the firm (ACCEL + proton therapy = Varian)
  - 455 long-lived large firms: the ones the LLM knows best

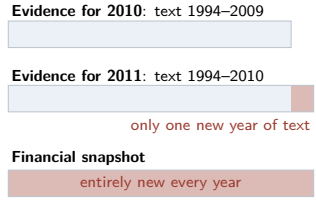


Suggestions

- Report the model and its cutoff; test on fiscal years after the cutoff
- Re-run with chronologically consistent LLMs (He et al. 2025; Yan et al. 2026)
- Probe memory: without retrieval, ask what firm X did after t (Sarkar & Vafa 2024)

Comment 1.3: Does new text drive within-firm changes in q^{AI}?

- Firm FE: identification comes from a firm's own changes over time
- But the text barely changes:
  - 10-Ks are mostly boilerplate (Cohen, Malloy & Nguyen 2020)
  - No time window or recency weighting: top-k keeps the most similar passages, so the same year-2000 passages may drive q^{AI} in 2010, 2011, even 2015
- $\rightarrow$ **The within-firm variation that identifies β may not come from new text**

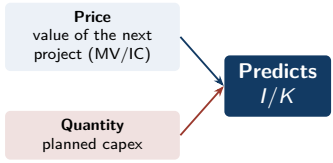


Suggestions

- Use rolling retrieval windows (e.g., $t-3$ to $t-1$)

Comment 2: Predicting investment $\neq$ measuring marginal q

- In q-theory, q is a **price**: the value of one more unit of capital
- Paper: q^{AI} predicts investment better than Q
- But any investment forecast (a **quantity**) would too (Gennaioli et al. 2016; Jha et al. 2024), and the inputs already contain one: capex guidance in calls, last year's capex in the snapshot
- $\rightarrow$ **q^{AI} may capture planned investment, not a shadow value**



Suggestions

- **Key test: ask the LLM to forecast I/K from the same evidence – does MV/IC add anything?**
- Valuation test: holding I/K fixed, does q^{AI} predict the profitability of new investment?

Conclusion

- This paper measures marginal q bottom-up: retrieval over firm text plus an LLM proposes each firm's next projects and values them
- q^{AI} predicts investment better than average Q, and the type of the marginal project shapes how investment responds
- The gap between average Q and q^{AI} traces to omitted intangibles and capitalized rents; prior fixes leave rents untouched
- My suggestions:
  - Show what drives q^{AI}: separate the text from the financial snapshot, the SFT anchor, and the LLM's memory
  - Show that q^{AI} is a price, not a quantity: compare it with a direct LLM forecast of investment

Thank you!